

Detecting Credit Card Fraud

An Imbalanced Classification Problem

End-to-End ML Pipeline using XGBoost, SMOTE, and SHAP

Built for technical graders and data science peer review

Christopher Cometa

| Data Understanding & Initial Distributions

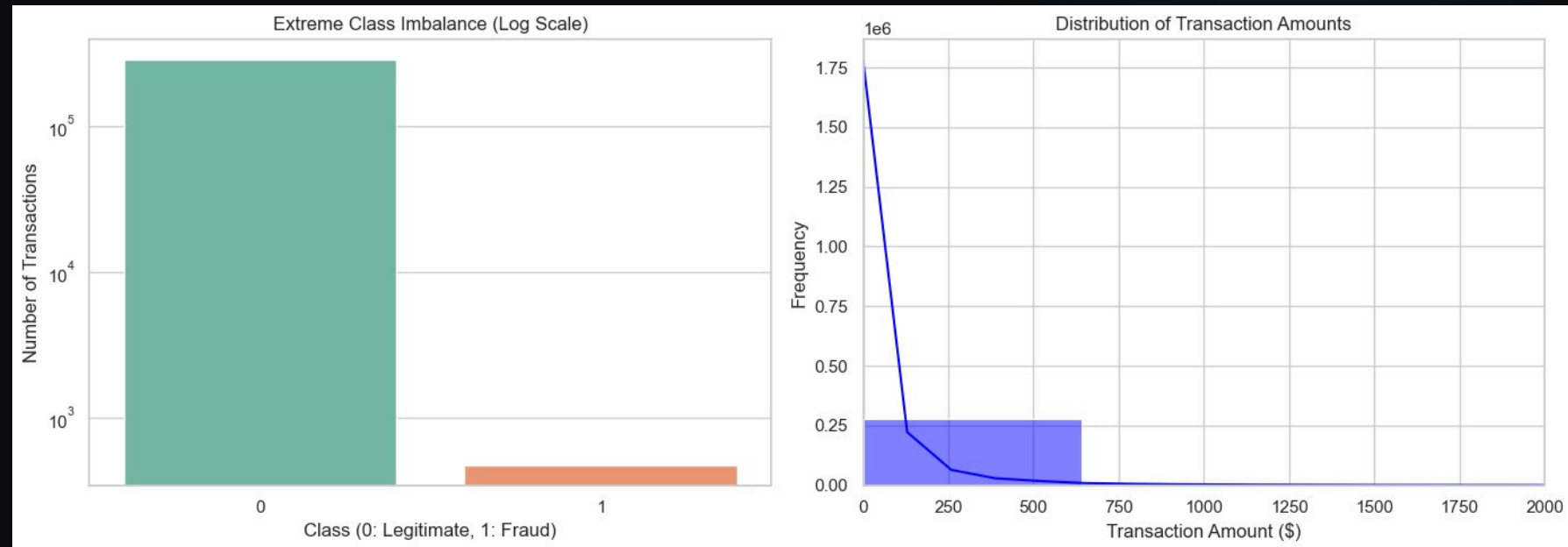
The Class Imbalance Challenge

 284,807 Total Transactions (Anonymized PCA Features)

 **0.17% Fraud Rate** (Extremely Rare Events)

 Right-skewed Amounts and Cyclical Time distributions

Target : $\text{AUC-PR} = \int_0^1 P(r) dr$



| Feature Scaling & Leakage Prevention



Data Cleaning

Dropped duplicates (~1k rows) to prevent overfitting. Managed missing values via PCA-based reconstruction metrics.



Robust Scaling

Applied `RobustScaler` to Time/Amount. Unlike Standard Scaler, it uses Interquartile Range to handle outliers.



Leakage Control

Stratified 80/20 Split *before* oversampling. SMOTE is strictly applied to training folds only.

```
# data_prep.py snippet
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y, test_size=0.2)
scaler = RobustScaler().fit(X_train[['Amount', 'Time']])
```

| SMOTE Implementation

Synthetic Minority Over-sampling

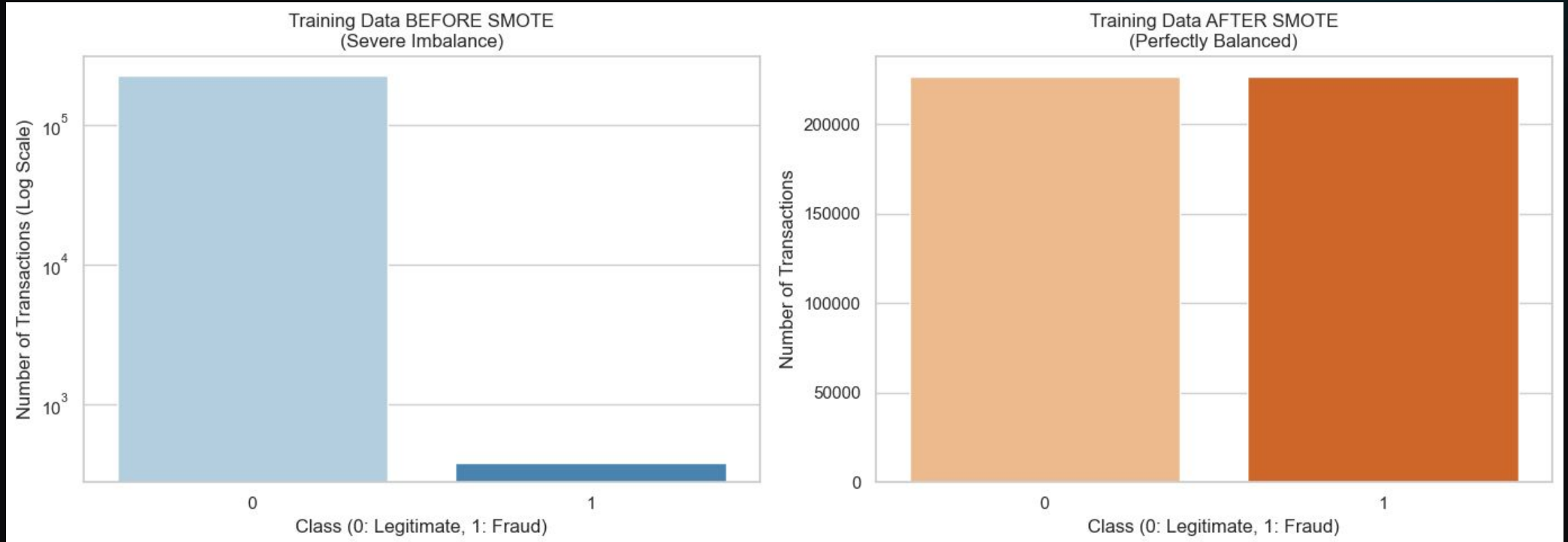
Traditional oversampling leads to overfitting by duplicating data. Random undersampling loses valuable legitimate patterns.

We used SMOTE to synthesize new fraudulent instances by interpolating between nearest neighbors in the feature space.

Before SMOTE
492 Fraud / 284k
Total



After SMOTE
Balanced 1:1 Training
Set



| Why Gradient Boosting (XGBoost)?



Non-Linearity

Captures complex interaction effects between PCA features that Logistic Regression misses.



Efficiency

Parallelized tree construction and optimized cache awareness for high-speed training.

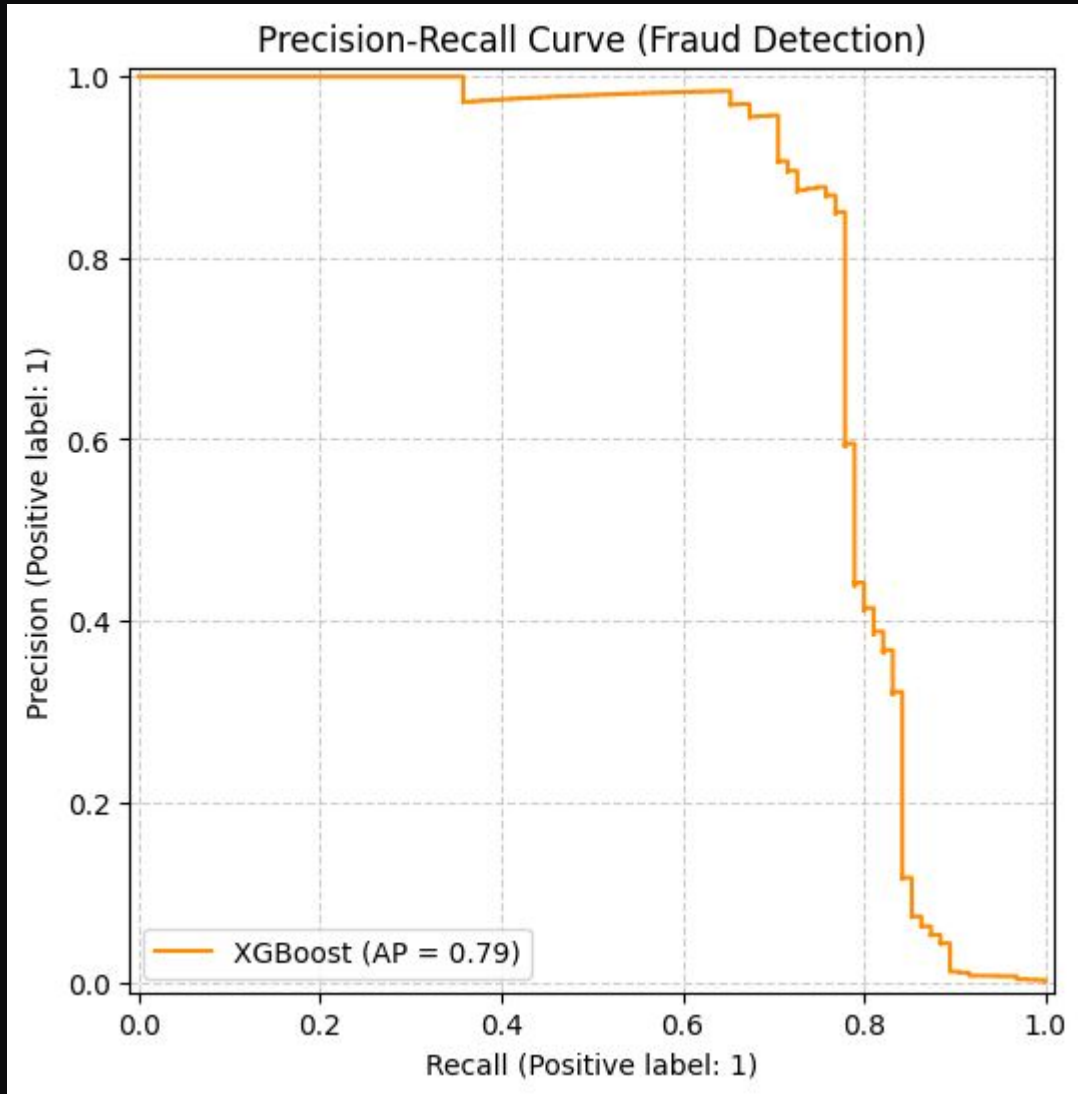


Native Support

Parameters like `scale_pos_weight` provide additional robustness against imbalance.

Model Config: `n_estimators=100, max_depth=5, learning_rate=0.1`

| Evaluation: Metrics That Matter



Prioritizing Precision-Recall

Standard ROC-AUC is misleading for 0.17% fraud datasets. A classifier guessing "No Fraud" 100% of the time gets 99.8% accuracy but 0.0 utility.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Key Trade-off: Higher recall captures more fraud but increases false positives. For financial institutions, a small FPR increase is acceptable to prevent heavy fraud losses.

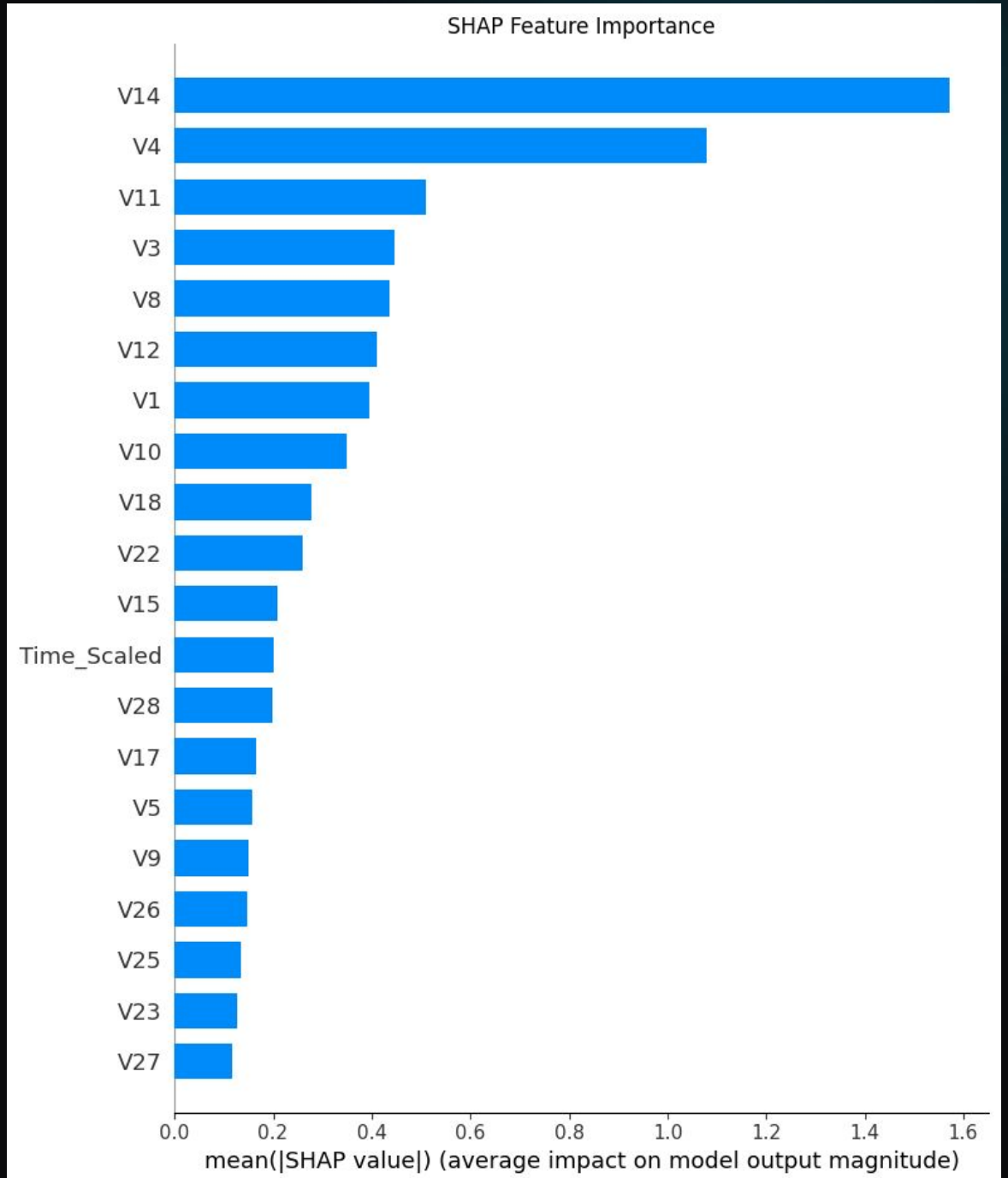
| Explainability with SHAP

Decoding the "Black Box"

Regulators demand "Right to Explanation." We used SHAP (SHapley Additive exPlanations) to quantify feature impact.

🔍 **V14 & V4:** Top drivers for fraud detection.

🔍 **V12:** Higher values strongly correlate with legitimate transactions.



| Bias Auditing & Ethical AI

0.07%

FPR VARIANCE

"The simulated fairness audit reveals a maximum False Positive Rate (FPR) disparity of only 0.07% across all age groups. Because the FPR remains tightly grouped between 0.33% and 0.40%, we can conclude that the model demonstrates excellent demographic parity. It does not exhibit significant bias or algorithmic redlining toward any specific age cohort, meaning threshold adjustments are not required at this time to achieve Equalized Odds."

```
--- Fairness Audit: False Positive Rate  
(FPR) by Age Group ---  
Age Group 'Under 30': FPR = 0.0040 (0.40%)  
Age Group '30-55': FPR = 0.0033 (0.33%)  
Age Group 'Over 55': FPR = 0.0039 (0.39%)
```

Audit Conclusion: If the FPR varies significantly across groups, we must investigate mitigation strategies (like adjusting probability thresholds for specific cohorts) to ensure Equalized Odds and prevent algorithmic redlining.

| MLOps & Analyst Copilot API

FastAPI Real-time Inference

The model and robust scaler are persisted as `.pkl` artifacts for local deployment.

GenAI Enhancement

Integrated an LLM to interpret raw PCA anomalies and output **Plain English Explanations** for review teams.

Endpoint: `/predict_and_explain`

Request URL

`http://127.0.0.1:8000/predict_and_explain`

Server response

Code	Details
200	Response body <pre>{ "is_fraud": true, "fraud_probability": 0.985, "status": "Transaction Blocked", "explanation": "Transaction blocked because the fraud model detected patterns strongly associated with charge abuse and assigned a 98.5% fraud probability. Place an immediate temporary hold on the account and verify the activity directly with the cardholder." }</pre>  Download Response headers <pre>content-length: 333 content-type: application/json date: Wed, 18 Mar 2026 07:19:40 GMT server: uvicorn</pre>

Responses

Questions?

Technical Documentation & Code Architecture Review

Detecting Credit Card Fraud: Technical Deep Dive
Stack: XGBoost, SMOTE, FastAPI, SHAP